Status	Finished
Started	Friday, 16 October 2026, 1:28 PM
Completed	Monday, 19 October 2026, 1:34 PM
Duration	3 days
Grade	**98.00** out of 100.00

Information

This is the last quiz for COSC264, covering the lecture material from the final two weeks of the course:

- TCP
- UDP
- Network applications
- Web and HTTP
- Email
- DNS

Question **1**

Correct

Mark 2.00 out of 2.00

A UDP socket is identified by:

Penalty regime: 33%, 66%, 100%

Select one or more:

- [x] destination IP address
- [x] destination port number
- [ ] source port number
- [ ] source IP address

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **2**

Correct

Mark 2.00 out of 2.00

A TCP socket is identified by:

Penalty regime: 33%, 66%, 100%

Select one or more:

- [x] destination port number
- [x] source port number
- [x] destination IP address
- [x] source IP address

Your answer is correct.

Correct

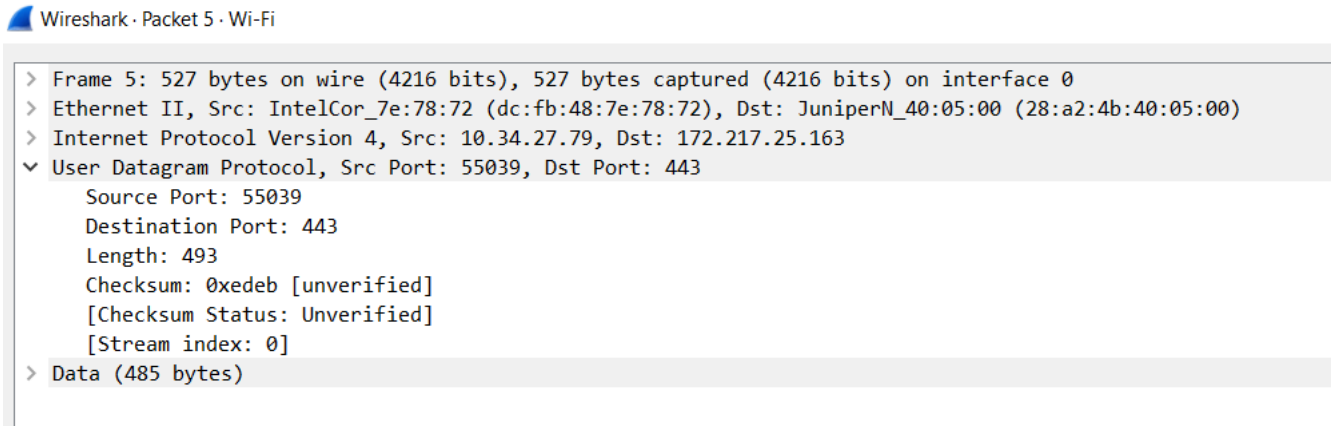
Marks for this submission: 2.00/2.00.

Question **3**

Correct

Mark 8.00 out of 8.00

A screenshot of a segment captured by WireShark is given below. Please fill in the blanks below so that the statements are true for this segment.



This is a [UDP] segment.

The source port number in this segment is [55039].

The destination port number in this segment is [443].

The length of this segment is [493] bytes; the size of the data field is [485] bytes. Therefore, the header occupies [8] bytes.

Penalty regime: 33%, 67%, 100%

Your answer is correct.

Correct

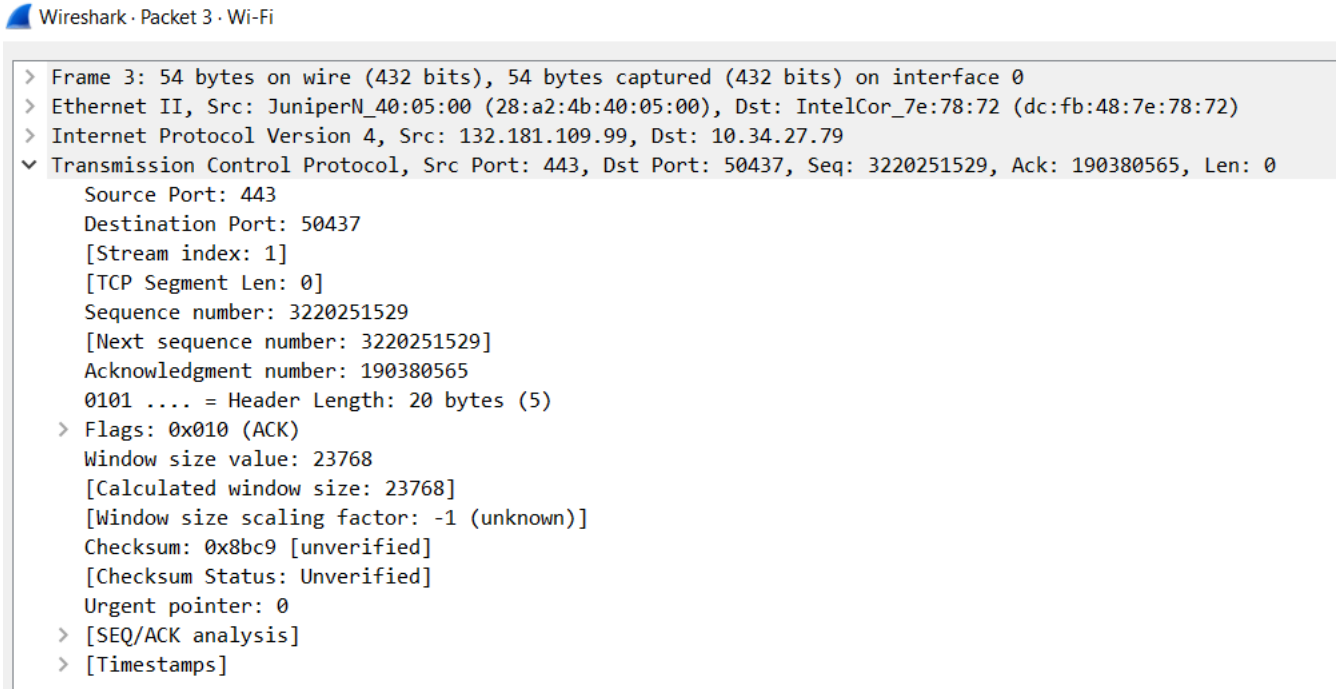
Marks for this submission: 8.00/8.00.

Question **4**

Correct

Mark 8.00 out of 8.00

A screenshot of a segment captured by WireShark is given below. Please fill in the blanks below so that the statements are true for this segment.



This is a TCP segment.

The source port number in this segment is 443.

The destination port number in this segment is 50437.

The sequence number in this segment is 3220251529.

The ACK number in this segment is 190380565.

The size of the receive window of this segment is 23768.

Penalty regime: 33%, 67%, 100%

Your answer is correct.

Correct

Marks for this submission: 8.00/8.00.

Question **5**

Correct

Mark 2.00 out of 2.00

In TCP 3-way handshaking, the client-side TCP first sends a SYN segment to the server-side TCP. Suppose the client randomly chooses an initial sequence number, *client_isn*. What is the ACK number in the corresponding SYNACK segment generated by the server?

Penalty regime: 33%, 66%, 100%

Select one:

- *client_isn +1*
- Another random number
- 0
- *client_isn*

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **6**

Correct

Mark 2.00 out of 2.00

Suppose the sequence number of the SYNACK segment (generated by the server) is *server_isn*. After receiving the SYNACK segment, the client sends back another segment acknowledging this reception. What is the sequence number of this segment?

Penalty regime: 33%, 66%, 100%

Select one:

- *client_isn*
- *client_isn +1*
- *server_isn*
- *server_isn +1*

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **7**

Correct

Mark 1.33 out of 2.00

Following the last question, what is the ACK number in the segment sent by the client after receiving the SYNACK segment from the server?

Penalty regime: 33%, 66%, 100%

Select one:

- *server_isn +1*
- *server_isn*
- *client_isn*
- *client_isn +1*

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00. Accounting for previous tries, this gives **1.33/2.00**.

Question **8**

Correct

Mark 2.00 out of 2.00

Which of the following statements are correct regarding TCP reliable data transfer service?

Penalty regime: 33%, 66%, 100%

Select one or more:

- When there is a timeout, the sender retransmits all not-yet-acknowledged segments.
- TCP usually uses cumulative acknowledgements.
- The reliable data transfer mechanism in TCP is different from both Go-back-N and Selective Repeat.
- Some segments can be retransmitted before timeout. This is called **fast retransmit**.

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **9**

Correct

Mark 2.00 out of 2.00

When TCP does the round-trip time sampling, it never computes a sample round-trip time (**SampleRTT**) for a segment that has been retransmitted. Why?

Penalty regime: 33%, 66%, 100%

Select one:

- A retransmitted segment is more likely to be corrupted.
- Retransmission can cause network congestion.
- If a sender retransmits a segment and receives its ACK, it does not know whether this ACK corresponds to the earlier segment or the retransmitted segment. The round-trip time estimation becomes inaccurate.
- A retransmitted segment is more likely to be delayed or lost again.

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **10**

Correct

Mark 8.00 out of 8.00

TCP uses a timeout/retransmit scheme to recover from lost segments. TCP needs to set the timeout value at least larger than the connection's round-trip time (RTT). TCP estimates the round-trip time based on the sampled RTTs. Upon obtaining a new sample RTT, TCP updates the **estimated RTT** according to the following formula:

$$\text{EstimatedRTT} = (1 - \alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}$$

That is, the new value of EstimatedRTT is a weighted combination of the previous value of EstimatedRTT and the new value of SampleRTT.

In addition to calculating the estimated RTT, TCP measures the variability of RTT, i.e., **DevRTT,** according the following formula:

$$\text{DevRTT} = (1 - \beta) * \text{DevRTT} + \beta * |\ \text{SampleRTT} - \text{EstimatedRTT}\ |$$

Here, β is a parameter and the operator |x| outputs the absolute value of x. (Hint: You can use `abs()` function in Python.)

Complete in the following Python 3 function so that it takes a *list* of sample RTT values and returns the last updated DevRTT after processing all the sample RTTs.

Notes:

- α = 0.125 and β = 0.25. These values have been pre-loaded into the answer box for you.
- The initial DevRTT is zero. Whenever a new sample RTT is read, you should calculate its estimated RTT and then use this to update DevRTT using the formula provided.
- Assume the first estimated RTT is equal to the first sample RTT. Susbequent estimated RTTs should be calculated based on the formula provided.

For example:

Test	Result
`print(f"{dev_rtt([1]):.3f}")`	`0.000`
`print(f"{dev_rtt([1,1,1]):.3f}")`	`0.000`
`print (f"{dev_rtt([1,2]):.3f}")`	`0.219`
`print (f"{dev_rtt([1,2,3]):.3f}")`	`0.574`

Answer: (penalty regime: 0, 10, 20, ... %)

Reset answer

```
def dev_rtt(sample_rtts):
    """Return the latest DevRTT based on the given list of sample RTTs."""
    alpha = 0.125
    beta = 0.25
    estimated_rtt = sample_rtts[0]
    dev = 0
    for i, sample in enumerate(sample_rtts):
        if i > 0:
            estimated_rtt = (1 - alpha) * estimated_rtt + alpha * sample
        dev = (1 - beta) * dev + beta * abs(sample - estimated_rtt)
    return dev
```

	Test	Expected	Got	
✓	`print(f"{dev_rtt([1]):.3f}")`	`0.000`	`0.000`	✓
✓	`print(f"{dev_rtt([1,1,1]):.3f}")`	`0.000`	`0.000`	✓
✓	`print (f"{dev_rtt([1,2]):.3f}")`	`0.219`	`0.219`	✓
✓	`print (f"{dev_rtt([1,2,3]):.3f}")`	`0.574`	`0.574`	✓
✓	`print (f"{dev_rtt([1,2,3,4]):.3f}")`	`1.008`	`1.008`	✓
✓	`print (f"{dev_rtt([1,2,3,4,5]):.3f}")`	`1.480`	`1.480`	✓
✓	`print(f"{dev_rtt([1,2,3,4,5,20]):.3f}")`	`5.025`	`5.025`	✓
✓	`print(f"{dev_rtt([39,93,40,44,40,39,39,39,39,40,80]):.3f}")`	`11.064`	`11.064`	✓

Passed all tests! ✓

Correct

Marks for this submission: 8.00/8.00.

Question **11**

Correct

Mark 8.00 out of 8.00

Now we can calculate the **timeout interval** for the TCP timer based on the following formula:

TimeoutInterval = EstimatedRTT + 4* DevRTT

Complete the following function so that it takes a list of sample RTTs and returns the last updated timeout interval. The values of EstimatedRTT and DevRTT should be calculated based on the instructions given in the previous question.

For example:

Test	Result
`print(f"{timeout_interval([1]):.3f}")`	`1.000`
`print(f"{timeout_interval([1, 1]):.3f}")`	`1.000`
`print(f"{timeout_interval([1, 2]):.3f}")`	`2.000`
`print(f"{timeout_interval([1, 2, 3]):.3f}")`	`3.656`

Answer: (penalty regime: 0, 10, 20, ... %)

Reset answer

```
def timeout_interval(sample_rtts):
    """Return the time interval based on the given list of sample RTTs."""
    alpha = 0.125
    beta = 0.25
    estimated_rtt = sample_rtts[0]
    dev = 0
    for i, sample in enumerate(sample_rtts):
        if i > 0:
            estimated_rtt = (1 - alpha) * estimated_rtt + alpha * sample
        dev = (1 - beta) * dev + beta * abs(sample - estimated_rtt)
    return estimated_rtt + 4 * dev
```

	Test	Expected	Got	
✓	`print(f"{timeout_interval([1]):.3f}")`	`1.000`	`1.000`	✓
✓	`print(f"{timeout_interval([1, 1]):.3f}")`	`1.000`	`1.000`	✓
✓	`print(f"{timeout_interval([1, 2]):.3f}")`	`2.000`	`2.000`	✓
✓	`print(f"{timeout_interval([1, 2, 3]):.3f}")`	`3.656`	`3.656`	✓
✓	`print(f"{timeout_interval([1, 2, 3, 4]):.3f}")`	`5.723`	`5.723`	✓
✓	`print(f"{timeout_interval([1, 2, 3, 4, 5]):.3f}")`	`8.025`	`8.025`	✓
✓	`print(f"{timeout_interval([1, 2, 3, 4, 5, 20]):.3f}")`	`24.441`	`24.441`	✓
✓	`print(f"{timeout_interval([39,93,40,44,40,39,39,39,39,39,40,80]):.3f}")`	`90.566`	`90.566`	✓

Passed all tests! ✓

Correct

Marks for this submission: 8.00/8.00.

Question **12**

Correct

Mark 1.33 out of 2.00

Which of the following statements are correct about TCP flow control?

Penalty regime: 33%, 66%, 100%

Select one or more:

- [ ] TCP flow control is the same as TCP congestion control.
- [x] TCP specification requires the sender to continue to send one-data-byte segments to its receiver even if the receiver's buffer is full.
- [x] TCP flow control is a speed-matching service.
- [x] In TCP flow control, the sender maintains a variable called **receive window** (RcvWindow) which indicates how much free buffer space is available at the receiver.

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00. Accounting for previous tries, this gives **1.33/2.00**.

Question **13**

Correct

Mark 9.00 out of 9.00

In the TCP Slow Start phase, the congestion window (CongWin) is set to an `initial` value and then doubles for every round trip (RTT).

Complete the following function `num_rtt(initial, ssthresh)` so that it outputs the number of round trips at which the congestion window exceeds the slow start threshold (`ssthresh`) for the first time.

Notes:

- The units for both `initial` and `ssthresh` are MSS (Maximum Segment Size).
- You may assume that `ssthresh` > `initial`.

For example:

Test	Result
`print(num_rtt(1, 16))`	`5`
`print(num_rtt(2, 16))`	`4`

Answer: (penalty regime: 0, 10, 20, ... %)

Reset answer

```
def num_rtt(initial, ssthresh):
    cwnd = initial
    rtt = 0
    while cwnd <= ssthresh:
        cwnd *= 2
        rtt += 1
    return rtt
```

	Test	Expected	Got	
✓	`print(num_rtt(1, 16))`	`5`	`5`	✓
✓	`print(num_rtt(2, 16))`	`4`	`4`	✓
✓	`print(num_rtt(3, 16))`	`3`	`3`	✓
✓	`print(num_rtt(2, 1234))`	`10`	`10`	✓

Passed all tests! ✓

Correct

Marks for this submission: 9.00/9.00.

Question **14**

Correct

Mark 2.00 out of 2.00

Which of the following applications need reliable data transfer?

Penalty regime: 33%, 66%, 100%

Select one or more:

- [x] Email
- [x] File transfer
- [ ] Live video streaming
- [x] Instant messaging

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **15**

Correct

Mark 2.00 out of 2.00

Which of the following applications have tight timing constraints on data delivery?

Penalty regime: 33%, 66%, 100%

Select one or more:

- [ ] File transfer
- [x] Teleconferencing
- [x] Internet telephony
- [ ] Email

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **16**

Correct

Mark 2.00 out of 2.00

In HTTP version 1.1, what is the first field below the request line?

Penalty regime: 33%, 66%, 100%

Select one:

- Connection.
- Any of these; HTTP is structured in plain-text. — Correct, unlike protocols like IP and TCP, HTTP does not structure its fields depending on their byte position in the packet. Rather, an HTTP packet is structured in plain-text with delimiters (sp, cr, and lf) separating header lines, field names, and values.
- Accept-language.
- User-agent.

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **17**

Correct

Mark 5.00 out of 5.00

Suppose you, as a server, receive the following HTTP message from a client browser:

```
GET /cosc264/index.html HTTP/1.1
Host: cosc.canterbury.ac.nz
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax)
Accept: ext/xml, application/xml, application/xhtml+xml, text/html;q=0.9, text/plain;q=0.8, image/png, */*;q=0.5
Accept-Language: en-gb;q=0.4, en-nz;q=0.5
Accept-Encoding: zip, deflate
Accept-Charset: ISO-8895-1, utf-8;q=0.7, *q=0
keep-Alive: 300
Connection: keep-alive
```

Please fill in the blanks below so that the statements are true for this message.

Penalty regime: 33%, 67%, 100%

This message is a [request] .

The preferred language is [en-nz] .

This message is part of a [persistent] connection.

Your answer is correct.

Correct

Marks for this submission: 5.00/5.00.

Question **18**

Correct

Mark 2.00 out of 2.00

HTTP is a stateless protocol. However, many sites wish to identify and track their users. Cookies are often used to track and add a state to HTTP. Where is the state information stored?

Penalty regime: 100%

Select one:

- Both client and server side
- Server side
- Client side

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **19**

Correct

Mark 2.00 out of 2.00

What are the benefits of web caching?

Penalty regime: 33%, 66%, 100%

Select one or more:

- [x] Reduces latency experienced by clients.
- [x] Lowers bandwidth costs.
- [x] Reduces network traffic.
- [ ] Requested data will be up to date.

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **20**

Correct

Mark 2.00 out of 2.00

Which of the following protocols is the principle application-layer protocol for Email?

Penalty regime: 33%, 66%, 100%

Select one:

- HTTP
- SMTP
- TCP
- UDP

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **21**

Correct

Mark 2.00 out of 2.00

Which transport-layer protocol is used by SMTP?

Penalty regime: 100%

Select one:

- TCP
- UDP

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **22**

Correct

Mark 2.00 out of 2.00

Which data format does SMTP accept?

Penalty regime: 33%, 66%, 100%

Select one:

- any non-ASCII
- base64
- 7-bit ASCII
- 8-bit ASCII

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **23**

Correct

Mark 2.00 out of 2.00

Which of the following mail access protocols can allow users to organise messages in folders?

Penalty regime: 100%

Select one:

- POP3
- IMAP

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **24**

Correct

Mark 2.00 out of 2.00

Which transport-layer protocol does DNS typically run over?

Penalty regime: 100%

Select one:

- UDP
- TCP

Your answer is correct.

Correct

Marks for this submission: 2.00/2.00.

Question **25**

Correct

Mark 1.33 out of 2.00

What are the main services provided by DNS?

Penalty regime: 33%, 66%, 100%

Select one or more:

- Hostname to IP translation
- Load distribution
- Host aliasing
- Mail server aliasing

Your answer is correct.

Correct

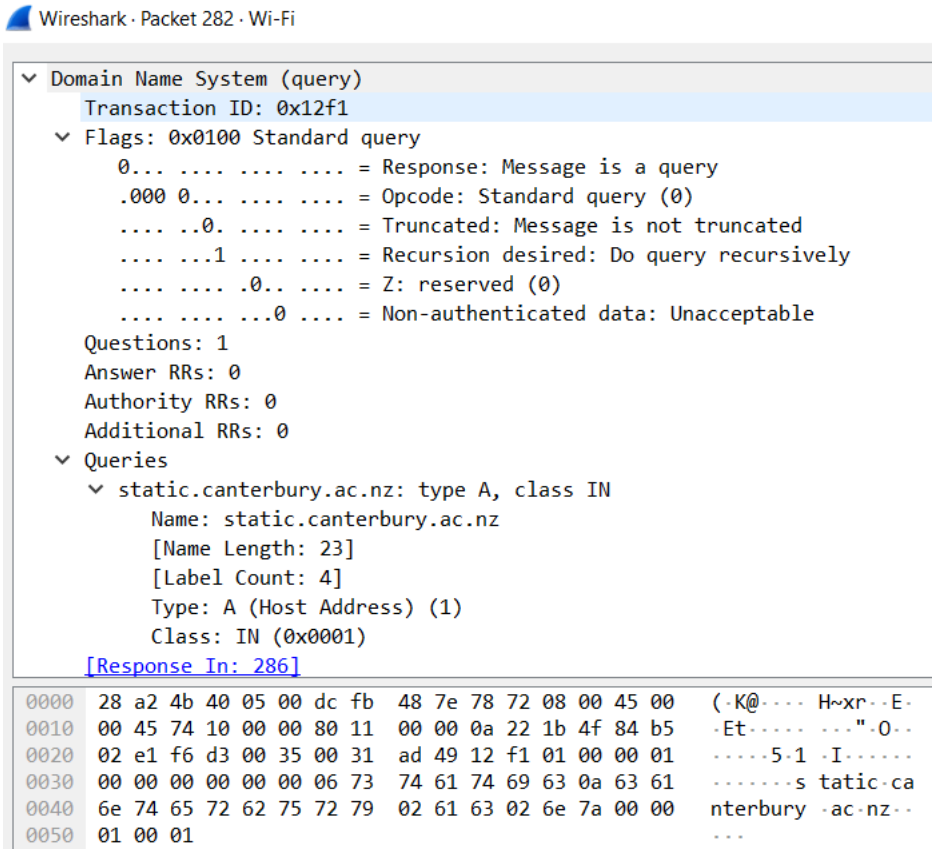
Marks for this submission: 2.00/2.00. Accounting for previous tries, this gives **1.33/2.00**.

Question **26**

Correct

Mark 5.00 out of 5.00

Below is a screenshot of a DNS message captured by Wireshark. Please fill in the blanks below so that the statements are true for this message.



This is a [query].

The type of DNS record being sought is [A].

The *Name* field for this DNS message is [static.canterbury.ac.nz].

Penalty regime: 33%, 67%, 100%

Your answer is correct.

Correct

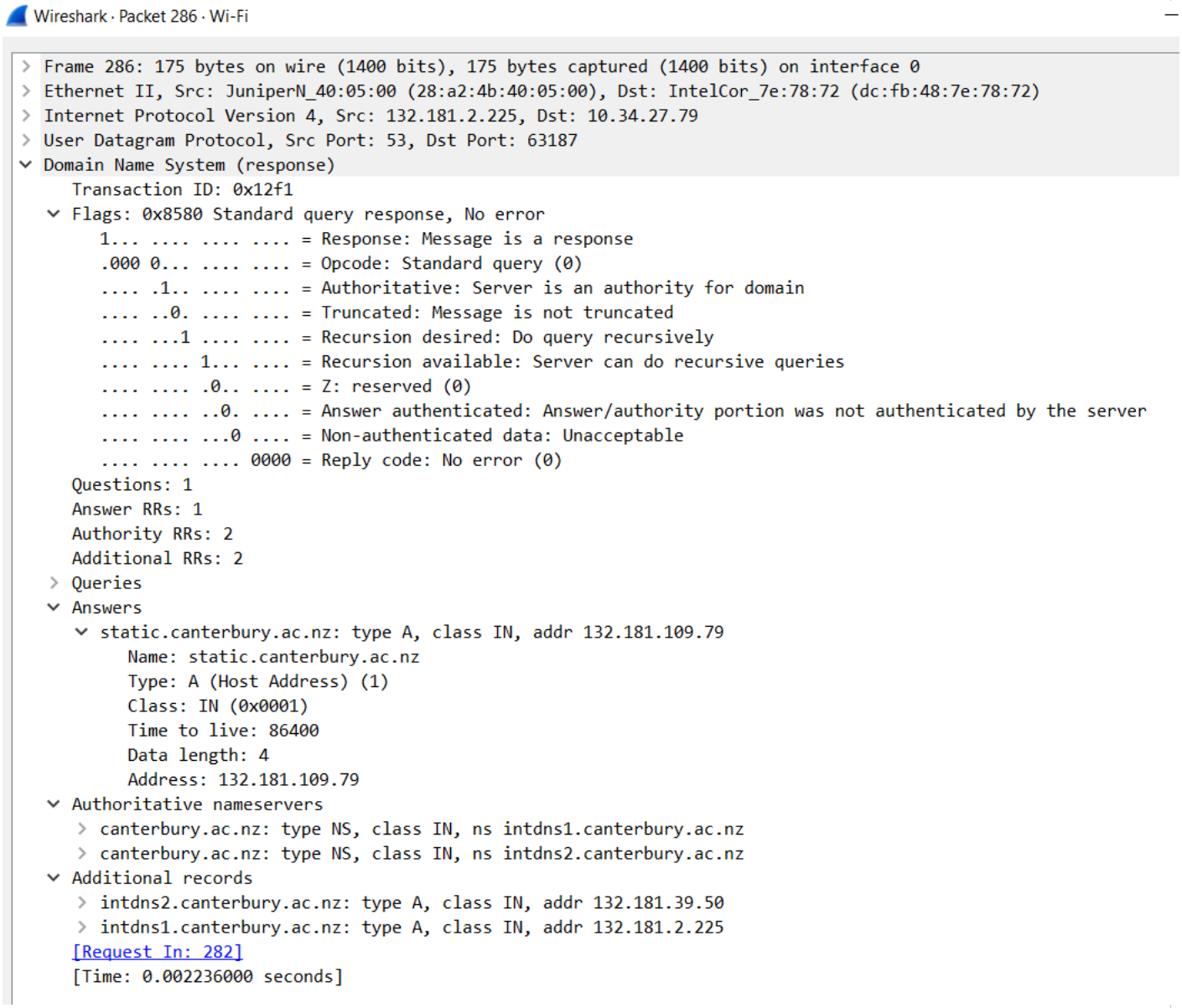
Marks for this submission: 5.00/5.00.

Question **27**

Correct

Mark 6.00 out of 6.00

Below is a screenshot of a DNS message captured by Wireshark. Please fill in the blanks below so that the statements are true for this message.



This is a [response] .

Based on this DNS message, we know that IP address of the host static.canterbury.ac.nz is [132.181.109.79] . (You can ping this host and check whether its IP has changed.)

Based on this message, there are [2] authoritative nameservers listed.

The IP address of the DNS nameserver "intdns2.canterbury.ac.nz" is [132.181.39.50] .

Penalty regime: 33%, 67%, 100%

Your answer is correct.

Correct

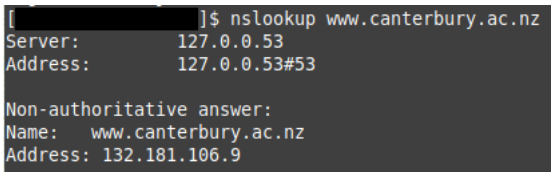
Marks for this submission: 6.00/6.00.

Question **28**

Correct

Mark 5.00 out of 5.00

The `nslookup` command enables you to use the Domain Name Service (DNS) to find the IP address of a host name. Below is a screenshot of the output for the command `nslookup www.canterbury.ac.nz`.



Skim the `man` page for `nslookup` (or search for an explanation online) and then match the following parts of the output to their meanings.

Penalty regime: 33%, 66%, 100%

`132.181.106.9`	IP address of www.canterbury.ac.nz
`127.0.0.53`	IP address of Domain Name Server (DNS)
`#53`	The port through which you communicate with the DNS server

Your answer is correct.

Correct

Marks for this submission: 5.00/5.00.

Question **29**

Not answered

Not graded

How did you find this quiz?

Note: This question is not worth any marks - we'd just like to know how you're doing.

- Very easy
- Easy
- Okay
- Hard
- Very hard

Thanks for your feedback.

Question **30**

Not answered

Not graded

Compared with the PowerPoint-based lecture slides used earlier in the course, how well did this week's web-based slides support your understanding of the material?

- a. Much worse than the PowerPoint-based slides
- b. Somewhat better than the PowerPoint-based slides
- c. Much better than the PowerPoint-based slides
- d. Somewhat worse than the PowerPoint-based slides
- e. About the same

Your answer is incorrect.

Question **31**

Not answered

Not graded

This question is not graded. How confident do you feel that you understood the main ideas from this week's lectures?

- a. Somewhat confident
- b. Very confident
- c. Somewhat unconfident
- d. Very unconfident
- e. Neither confident nor unconfident

Your answer is incorrect.

Information

Thanks for sticking with us to the end.

If you haven't already, please read the exam information in the *Test and Exam Information* section.

Best wishes for the final exam.

- The COSC264 teaching team